

## <2018학년도 수학1 기말고사 풀이>

1.(1) 40

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\ &= 5 \int_0^{2\pi} \sqrt{2(1-\cos\theta)} d\theta = 10 \int_0^{2\pi} \sin \frac{\theta}{2} d\theta = 40 \end{aligned}$$

(2)

수평인 점  $(5\pi, 10)$ , 수직인 점  $(0, 0), (10\pi, 0)$

$$\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\text{수평} \Leftrightarrow \frac{dy}{dx} = 0 \Leftrightarrow \sin \theta = 0, \cos \theta \neq 1 \therefore \theta = \pi, (5\pi, 10)$$

$$\text{수직} \Leftrightarrow \frac{dy}{dx} = \pm \infty, \lim_{\theta \rightarrow 2n\pi^-} \frac{\sin \theta}{1 - \cos \theta} = \lim_{\theta \rightarrow 2n\pi^+} \frac{\cos \theta}{\sin \theta} = \infty \quad (\because \frac{0}{0}, \text{로피탈의 정리})$$

$$\text{비슷하게} \lim_{\theta \rightarrow 2n\pi^-} \frac{dy}{dx} = -\infty \therefore \theta = 0, 2\pi \quad (0, 0), (10\pi, 0)$$

(3)

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{d\theta} \left( \frac{dy}{dx} \right)}{\frac{dx}{d\theta}} = \frac{-1}{5(1 - \cos \theta)^2}$$

$\theta = \frac{\pi}{3}$  를 대입하여 계산하면  $-\frac{4}{5}$  이다.

2.

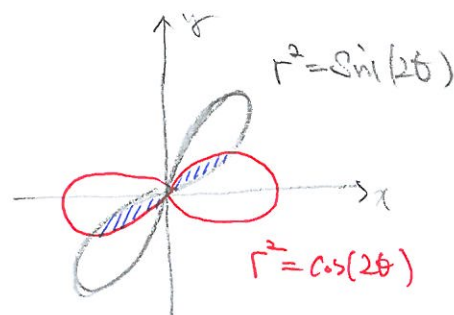
$$\tan^{-1}x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad |x| < 1$$

$$x = \frac{1}{\sqrt{3}}, \quad \frac{\pi}{6} = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{\sqrt{3}}\right)^{2n+1}}{(2n+1)} = \frac{1}{\sqrt{3}} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n}$$

$$\Rightarrow \pi = 2\sqrt{3} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)3^n}$$

3.(1)

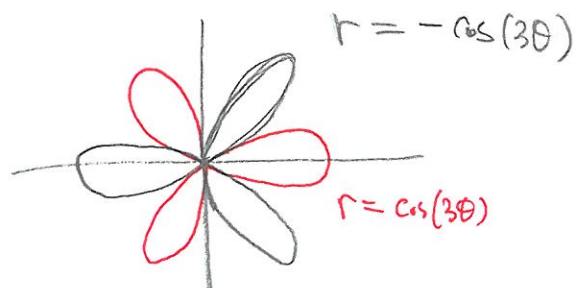
$$\begin{aligned} A &= 2 \left\{ \int_0^{\frac{\pi}{8}} \frac{1}{2} \sin 2\theta \, d\theta + \int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \frac{1}{2} \cos 2\theta \, d\theta \right\} \\ &= \frac{1}{2} - \frac{\sqrt{2}}{4} + \left( \frac{1}{2} - \frac{\sqrt{2}}{4} \right) = 1 - \frac{\sqrt{2}}{2} \end{aligned}$$



(2)

$$r = |\cos(3\theta)| \Leftrightarrow \pm r = \cos(3\theta)$$

$$\begin{aligned} A &= 12 \int_0^{\frac{\pi}{6}} \frac{1}{2} \cos^2(3\theta) \, d\theta \\ &= 6 \int_0^{\frac{\pi}{6}} \frac{1}{2} \{\cos(6\theta) + 1\} \, d\theta \\ &= 3 \cdot \frac{\pi}{6} = \frac{\pi}{2} \end{aligned}$$



4.(1)

①  $a_1 = 1 \leq (\sqrt{2})^1 = a_2$

②  $n = k$  일 때  $a_k \leq a_{k+1}$  라고 가정하자.

$$n = k+1 \text{ 이면 } a_{k+1} = \sqrt{2} a_k \leq \sqrt{2} a_{k+1} = a_{k+2}$$

①과 ②에 의해 모든  $n$ 에 대해  $a_n \leq a_{n+1}$  을 만족한다.

$\therefore \{a_n\}$  는 단조수열이다.

(2)

모든  $n$ 에 대해 (1)에 의해  $1 = a_1 \leq a_n$  이다.

$a_k \leq 2$  라 가정하자.

$$a_{k+1} = \sqrt{2} a_k \leq (\sqrt{2})^2 = 2$$

수학적 귀납법에 의해  $a_n \leq 2$  이다.

5.(1)

$$n \geq 30 > 27 > 3^3 > e^e$$

$$\ln(\ln e^e) = \ln e = 1 > 0$$

$$(\ln(\ln n))^{\ln n} = (e^{\ln(\ln(\ln n))})^{\ln n} = (e^{\ln n})^{\ln(\ln(\ln n))} = n^{\ln(\ln(\ln n))}$$

충분히 큰 수  $n$ 에 대해  $\ln(\ln(\ln n)) \rightarrow \infty$  이고

$$\ln(\ln(\ln n)) > 2 \text{ 이므로}$$

$$(\ln(\ln n))^{\ln n} = n^{\ln(\ln(\ln n))} \geq n^2$$

$$\Rightarrow \frac{1}{(\ln(\ln n))^{\ln n}} \leq \frac{1}{n^2} \quad (\text{충분히 큰 } n \text{에 대해})$$

$\sum \frac{1}{n^2}$  이 수렴하므로 비교판정법에 의해  $\sum_{n=30}^{\infty} \frac{1}{(\ln(\ln n))^{\ln n}}$  : 수렴

$$\sum_{n=30}^{\infty} \left| \frac{1}{(\ln(\ln n))^{\ln n}} \right| = \sum_{n=30}^{\infty} \frac{1}{(\ln(\ln n))^{\ln n}} \text{ 이므로 절대수렴한다.}$$

(2)

$$\left| \ln \left( \cos \frac{1}{n} \right) \right| = -\ln \left( \cos \frac{1}{n} \right)$$

$a_n = -\ln \left( \cos \frac{1}{n} \right), b_n = \frac{1}{n^2}$  이라 하면  $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$  은 양항급수이고

$$\lim_{n \rightarrow \infty} \frac{-\ln \left( \cos \frac{1}{n} \right)}{\frac{1}{n^2}} = \lim_{t \rightarrow 0} \frac{-\ln(\cos t)}{t^2} = \lim_{t \rightarrow 0} \frac{-\frac{-\sin t}{\cos t}}{2t} = \lim_{t \rightarrow 0} \frac{\tan t}{2t} = \frac{1}{2} > 0$$

$\sum_{n=1}^{\infty} \frac{1}{n^2}$  이 수렴하므로 극한비교판정법에 의해

$\sum_{n=1}^{\infty} \left| \ln \left( \cos \frac{1}{n} \right) \right|$  이 수렴한다.

$\therefore \sum_{n=1}^{\infty} \ln \left( \cos \frac{1}{n} \right)$  : 절대수렴

(3)

$$\textcircled{1} \sum_{n=2}^{\infty} \left| (-1)^n \frac{\tan^{-1} n}{\sqrt{n}} \right| = \sum_{n=2}^{\infty} \frac{\tan^{-1} n}{\sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{\tan^{-1} n}{\sqrt{n}}}{\frac{1}{\sqrt{n}}} = \frac{\pi}{2} > 0 \text{ 이므로 극한비교판정법에 의해}$$

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$  이 발산하므로  $\sum_{n=2}^{\infty} \frac{\tan^{-1} n}{\sqrt{n}}$  도 발산한다.

$$\textcircled{2} \sum_{n=2}^{\infty} (-1)^n \frac{\tan^{-1} n}{\sqrt{n}} : \text{교대급수}$$

$$b_n = \frac{\tan^{-1} n}{\sqrt{n}} \text{ 이라 하자}$$

$$\Rightarrow \bullet b_n \geq 0 \quad (n \geq 2)$$

$$\bullet \{b_n\} : \text{감소수열} \quad (n \geq 2)$$

$$(\because f(x) = \frac{\tan^{-1} x}{\sqrt{x}} \text{ 라 하면})$$

$$f'(x) = \frac{2x - \tan^{-1} x (1+x^2)}{2x \sqrt{x} (1+x^2)} \leq \frac{2x - (1+x^2)}{2x \sqrt{x} (1+x^2)} < 0 \quad (x \geq 2)$$

$$\bullet \lim_{n \rightarrow \infty} b_n = 0$$

$$(\because 0 \leq \frac{\tan^{-1} n}{\sqrt{n}} \leq \frac{\pi}{2} \cdot \frac{1}{\sqrt{n}} \text{ 이고 } \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0 \text{ 이므로})$$

$$\text{조임정리에 의해 } \lim_{n \rightarrow \infty} \frac{\tan^{-1} n}{\sqrt{n}} = 0)$$

그러므로 교대급수판정법에 의해 급수는 수렴한다.

$$\therefore \textcircled{1} \text{과 } \textcircled{2} \text{에 의해 } \sum_{n=2}^{\infty} (-1)^n \frac{\tan^{-1} n}{\sqrt{n}} \text{ 는 조건수렴한다.}$$

6.(1)

$$\sum_{n=2}^{\infty} (-1)^n \frac{(2x-1)^n}{4^n \ln n} : \text{수렴반지름, 수렴구간 구하기}$$

$$a_n = (-1)^n \frac{(2x-1)^n}{4^n \cdot \ln n}, \quad \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{|2x-1|}{4}$$

$$\therefore \text{수렴반지름} : 2$$

$$x = -\frac{3}{2}, \sum_{n=2}^{\infty} \frac{1}{\ln n} : \text{발산} \quad (\because 0 \leq \frac{1}{n} \leq \frac{1}{\ln n}, \sum_{n=2}^{\infty} \frac{1}{n} : \text{발산})$$

$$x = \frac{5}{2}, \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n} : \text{수렴} \quad (\because \text{교대급수판정법})$$

$$\therefore \text{수렴구간} : -\frac{3}{2} < x \leq \frac{5}{2}$$

(2)

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad |x| < 1$$

$$\Rightarrow \ln(1+x^2) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{n}, \quad |x^2| < 1 \Rightarrow |x| < 1$$

$$\begin{aligned} \int_0^{\frac{1}{2}} \ln(1+x^2) dx &= \int_0^{\frac{1}{2}} \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{n} dx \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n(2n+1)2^{2n+1}} \\ &= \frac{1}{3 \cdot 2^3} - \frac{1}{2 \cdot 5 \cdot 2^5} + \frac{1}{3 \cdot 7 \cdot 2^7} - \dots : \text{교대급수} \\ \frac{1}{2 \cdot 5 \cdot 2^5} &= \frac{1}{320} > \frac{1}{2000}, \quad \frac{1}{3 \cdot 7 \cdot 2^7} = \frac{1}{2688} < \frac{1}{2000} \\ \int_0^{\frac{1}{2}} \ln(1+x^2) dx &\approx \frac{1}{3 \cdot 2^3} - \frac{1}{2 \cdot 5 \cdot 2^5} = \frac{37}{960} (\text{교대급수 추정정리}) \end{aligned}$$

7.(1-1)

$$\tan^{-1}\left(x - \frac{1}{2}\right) = \left(x - \frac{1}{2}\right) - \frac{1}{3}\left(x - \frac{1}{2}\right)^3 + \frac{1}{5}\left(x - \frac{1}{2}\right)^5 + \dots$$

$$\frac{1}{2-2x} = \frac{1}{1-2(x-1/2)} = 1 + 2\left(x - \frac{1}{2}\right) + 2^2\left(x - \frac{1}{2}\right)^2 + 2^3\left(x - \frac{1}{2}\right)^3 + \dots$$

$$\begin{aligned} &\tan^{-1}\left(x - \frac{1}{2}\right) \left(\frac{1}{2-2x}\right) \\ &= \left\{ \left(x - \frac{1}{2}\right) - \frac{1}{3}\left(x - \frac{1}{2}\right)^3 + \frac{1}{5}\left(x - \frac{1}{2}\right)^5 + \dots \right\} \left\{ 1 + 2\left(x - \frac{1}{2}\right) + 2^2\left(x - \frac{1}{2}\right)^2 + 2^3\left(x - \frac{1}{2}\right)^3 + \dots \right\} \\ &= \left(x - \frac{1}{2}\right) + 2\left(x - \frac{1}{2}\right)^2 + \frac{11}{3}\left(x - \frac{1}{2}\right)^3 + \dots \end{aligned}$$

$$T_3(x) = \left(x - \frac{1}{2}\right) + 2\left(x - \frac{1}{2}\right)^2 + \frac{11}{3}\left(x - \frac{1}{2}\right)^3$$

(1-2)

$$f^{(3)}\left(\frac{1}{2}\right) = 3! \frac{11}{3} = 22$$

(2)

$$f(x) = \sqrt[5]{1+x}, f'(x) = \frac{1}{5}(1+x)^{-\frac{4}{5}}, f''(x) = -\frac{4}{25}(1+x)^{-\frac{29}{5}}$$

$$|f''(x)| \leq \frac{4}{25} = M, 0 < x \leq 0.01 \text{ 이므로}$$

$$\text{오차의 한계는 } |R_1(x)| \leq \frac{1}{2} \left( \frac{4}{25} \right) \left( \frac{1}{100} \right) = 8 \cdot 10^{-6} \text{ 이다.}$$

$$\sqrt[5]{1+x} \approx 1 + \frac{x}{5}, 0 < x \leq 0.01$$

(3)

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p} \quad p > 1 \text{ 이면 수렴}$$

$$0 < p \leq 1 \text{ 이면 발산}$$

$$f(x) = \frac{1}{x(\ln x)^p}$$

i)  $[2, \infty)$  에서 양함수, 연속함수

$$\text{ii) } f'(x) = -\frac{p + \ln x}{x^2(\ln x)^{p+1}} < 0 \Leftrightarrow x > e^{-p}, p > 0$$

$$\begin{aligned} \textcircled{1} \quad p=1 \text{ 일 때 } \int_2^{\infty} \frac{1}{x \ln x} dx &= \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x \ln x} dx \\ &= \lim_{t \rightarrow \infty} \ln(\ln t) - \ln(\ln 2) = \infty \text{ 발산} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad p \neq 1 \text{ 일 때 } \int_2^{\infty} \frac{1}{x(\ln x)^p} dx &= \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x(\ln x)^p} dx \\ &= \lim_{t \rightarrow \infty} -\frac{1}{p-1} (\ln t)^{1-p} - (\ln 2)^{1-p} \\ &= \begin{cases} -\frac{(\ln 2)^{1-p}}{1-p} : \text{수렴} & (p > 1) \\ \infty & (p < 1) \end{cases} \end{aligned}$$

8.(1)

$\forall |x| \leq 1, |\sin x| \leq |x|$  이고,  $|a| < 1$  이므로,

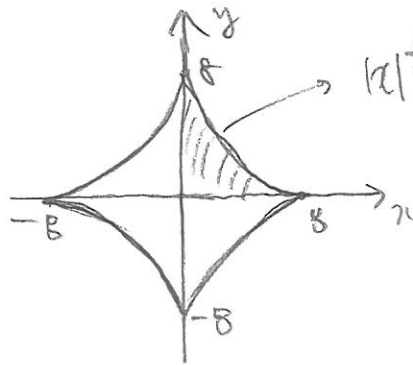
$$0 \leq \left| (-1)^n n! \sin a \cdot \sin\left(\frac{a}{2}\right) \cdots \sin\left(\frac{a}{n}\right) \right| \leq n! \cdot |a| \cdot \frac{|a|}{2} \cdots \frac{|a|}{n} = |a|^n, \forall n$$

$$\sum_{n=1}^{\infty} |a|^n = \frac{|a|}{1-|a|} \quad : \text{수렴}$$

$$\text{비교판정법, } \sum_{n=1}^{\infty} \left| (-1)^n n! \sin a \cdot \sin\left(\frac{a}{2}\right) \cdots \sin\left(\frac{a}{n}\right) \right| : \text{수렴}$$

$$\therefore \sum_{n=1}^{\infty} (-1)^n n! \sin a \cdot \sin \frac{a}{2} \cdots \sin \frac{a}{n} : \text{절대수렴}$$

(2)



$$|x|^{2/3} + |y|^{2/3} = 4$$

$$\begin{cases} x = 8 \cos^3 \theta \\ y = 8 \sin^3 \theta \end{cases}$$

$$S = 4 \int_0^8 y dx = 4 \int_{\pi/2}^0 8 \sin^3 \theta \cdot 8 \cdot 3 \cos^2 \theta (-\sin \theta) d\theta$$

$$(\because x = 8 \cos^3 \theta, dx = 8 \cdot 3 \cos^2 \theta (-\sin \theta) d\theta)$$

$$= 4 \cdot 8^2 \cdot 3 \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta = 4 \cdot 8^2 \cdot 3 \cdot \frac{\pi}{32} = 24\pi$$

$$(\because \sin^4 \theta \cos^2 \theta = \frac{1 - \cos 2\theta}{2} \cdot \frac{\sin^2 2\theta}{2} = \frac{1}{8} (1 - \cos 2\theta) \sin^2 2\theta)$$